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Q1. Program based on Classes and Objects - Initializaing Data member using Objects directly

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class student{
    int rollno;
    String name;
    double marks;
    void displayDetails(){
        System.out.println("Roll No: "+ rollno);
        System.out.println("Name: "+ name);
        System.out.println("Marks: "+marks);
    }

   public static void main(String[] args){
       Scanner sc=new Scanner(System.in);
       student s=new student();
       s.rollno=sc.nextInt();
       sc.nextLine();
       s.name=sc.nextLine();
       s.marks=sc.nextDouble();
       s.displayDetails();
   }
}
```

INPUT

```
192372182
sumanth
90.00
```

OUTPUT

```
Roll No: 192372182
Name: sumanth
Marks: 90.0
```

Q2. Classes and Objects-Initialize Object Data Using Setter Methods

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class student{
    private int rollno;
    private String name;
    private double marks;
    public void setRollNo(int rollno){
        this.rollno=rollno;
    }
    public void setName(String name){
        this.name=name;
    }
    public void setMarks(double marks){
        this.marks=marks;
    }
    public void displayDetails(){
        System.out.println("Roll No: "+rollno);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }
    public static void main(String[] args){
        Scanner sc =new Scanner(System.in);
        student s=new student();
        s.setRollNo(sc.nextInt());
        sc.nextLine();
        s.setName(sc.nextLine());
        s.setMarks(sc.nextDouble());
        s.displayDetails();
    }
}
```

INPUT

```
192372182
sumanth
90.00
```

OUTPUT

```
Roll No: 192372182
Name: sumanth
Marks: 90.0
```

Q3. Student Details Using Default and Parameterized Constructors

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Student{
    int rollNo;
    String name;
    double marks;
    Student(){
        rollNo=101;
        name="Rahul";
        marks=85.5;
    }
    Student(int rollNo,String name,double marks){
        this.rollNo=rollNo;
        this.name=name;
        this.marks=marks;
    }
    void displayDetails(){
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        Student s1=new Student();
        s1.displayDetails();
        int rollNo=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double marks=sc.nextDouble();
        Student s2=new Student(rollNo,name,marks);
        s2.displayDetails();
        sc.close();
    }
}
```

INPUT

```
192372182
sumanth
90.00
```

OUTPUT

```
Roll No: 101
Name: Rahul
Marks: 85.5
Roll No: 192372182
Name: sumanth
Marks: 90.0
```

Q4. Net Salary Calculation Using Single Inheritance

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class employee{
    int empid;
    String empname;
    double salary;
    void getemp(int empid,String empname,double salary){
        this.empid=empid;
        this.empname=empname;
        this.salary=salary;
    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        int empid=sc.nextInt();
        sc.nextLine();
        String empname=sc.nextLine();
        double salary=sc.nextDouble();
        Salary s=new Salary();
        s.getemp(empid,empname,salary);
        s.calculatesalary();
        s.display();
    }
}
class Salary extends employee{
    double hra,da,netsalary;
    void calculatesalary(){
        hra=salary*0.20;
        da=salary*0.10;
        netsalary=salary+hra+da;
    }
    void display(){
        System.out.println("Employee ID: "+empid);
        System.out.println("Employee Name: "+empname);
        System.out.println("Basic Salary: "+salary);
        System.out.println("HRA: "+ hra);
        System.out.println("DA: "+ da);
        System.out.println("Net Salary: "+netsalary);
    }
}
```

INPUT

```
123
sumanth
2000000
```

OUTPUT

```
Employee ID: 123
Employee Name: sumanth
Basic Salary: 2000000.0
HRA: 400000.0
DA: 200000.0
Net Salary: 2600000.0
```

Q5. Employee Gross Salary Calculation Using Multilevel Inheritance

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class employee{
    int empid;
    String empname;
    void emp(int empid,String empname){
        this.empid=empid;
        this.empname=empname;
    }
    public static void main(String[] args){
       Scanner sc=new Scanner(System.in);
        int empid=sc.nextInt();
        sc.nextLine();
        String empname=sc.nextLine();
        double basicsal=sc.nextDouble();
        gsal g=new gsal();
        g.emp(empid,empname);
        g.getsalary(basicsal);
        g.cal();
        g.display();
        sc.close();
    }
}
class salary extends employee{
    double basicsal;
    void getsalary(double basicsal){
        this.basicsal=basicsal;
    }
}
class gsal extends salary{
    double hra,da,grosssal;
    void cal(){
        hra=basicsal*0.20;
        da=basicsal*0.10;
        grosssal=basicsal+hra+da;
    }
    void display(){
        System.out.println("Employee ID: "+empid);
        System.out.println("Employee Name: "+empname);
        System.out.println("Basic Salary: "+basicsal);
        System.out.println("HRA: "+hra);
        System.out.println("DA: "+da);
        System.out.println("Gross Salary: "+grosssal);
    }
}
```

INPUT

```
123
sumanth
2000000
```

OUTPUT

```
Employee ID: 123
Employee Name: sumanth
Basic Salary: 2000000.0
HRA: 400000.0
DA: 200000.0
Gross Salary: 2600000.0
```